

Cleaning the flat-panel screens

Flat-panel monitors and screens making use of technologies like LCD, plasma, OLED, field-emission and nano-emission are becoming a part of our daily life in homes and offices with various electronic devices like computers, laptops, TVs and mobiles.

For the best viewing results, it is important to keep the screen clean from dust, dirt and fingerprints. This helps to reduce glare and produce clearer picture. Flat screens (LCD and plasma) are a little different than their CRT cousins when it comes to cleaning. The standard CRT monitor has a glass screen and can be cleaned in pretty much the same way as we would any other glass in our home or office. Flat screens, however, require a bit of special care when cleaning, as these are much more sensitive and get easily scratched and damaged. You need to take precautions when cleaning, lighting or covering these displays.

To avoid scratches or degeneration of the screen quality, observe the following instructions:

1. Use a protective computer filter to minimise the opportunity for dirt or damage to the screen.
2. Read the manufacturer's cleaning suggestions carefully to implement them.
3. Prior to cleaning your unit, unplug it.
4. Use a soft cloth such as soft, untreated eyeglass cloth or micro-fibre cleaning cloth, rather than tissues. Avoid using paper towels, toilet paper, tissue paper, or something like your shirt to wipe the flat screens. These non-ultra-soft materials can easily scratch the screen.
5. Use correct solution for cleaning. Water or an isopropyl alcohol based screen cleaner can be used for the purpose.
6. If necessary, dampen the cloth with distilled water or a mixture having equal parts of distilled water and white vinegar.
7. Never use strong cleaners as they may damage the anti-glare coating found on most screen displays. Avoid cleaning products that contain ammonia, ethyl alcohol, acetone, toluene, ethyl acid or methyl chloride. These chemicals can react with the materials that the flat screen is made of and yellow the screen or cause other kinds of damage.
8. Spray the cloth, not the screen. That is, do not spray the liquid directly onto the screen.
9. Gently wipe the screen in a consistent motion, such as counter clockwise, rather than haphazardly.
10. Use the cleaner sparingly to avoid leakage of excess fluid into the keypad.
11. Only clean the enclosure of the unit from the outside. Never open the casing—it may risk electrical shock besides causing possible damage to the electronics inside.
12. If the dry cloth doesn't completely clean the screen, do not press harder in an attempt to scrub the dirt off. Pushing directly on the flat screen can often cause pixels to burn out.
13. The plastic edge that surrounds the screen can be cleaned with any multipurpose cleaner but take care to avoid contact with the screen itself.

and require acceleration energies of a similar level. However, these eliminate much of the electrical complexity of a CRT, including heated filaments in the electron gun and electromagnets used to steer the beam. FEDs are thus much more power-efficient than a CRT of the same size.

LCDs, the most common flat-screen displays of today, use a bright light source at the back of the screen and then filter out most of the light to produce red, green and blue (RGB) sources for the sub-pixels. That means only one-third (or less in practice) of the light being generated at the back of the tube reaches the screen, at best. In most cases, the LCD itself then filters out additional light in order to change the brightness of the sub-pixels and produce a colour gamut. So, in spite of using extremely efficient light sources like cold-cathode fluorescent lamps or high-power white LEDs, the overall efficiency of an LCD is not very high.

To date, however, manufacturing problems have prevented FED systems from entering commercial production.

SEDs use a more stable method

FEDs are closely related to another developing display technology, the surface-conduction electron-emitter display (SED). These use a more stable but less efficient method to achieve a similar look to nano-emissive display (NED)—Motorola's term for their carbon-nanotube-based FED technology. The SED uses only one cathode ray tube, as opposed to more unre-



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